

PAPER_1590 — Rydberg $R_\infty \approx R_\infty = F \cdot SO_5 + F \cdot SSQ + F^2 \cdot D_{phys} = 1.0974$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum/Atomic **Date:** June 18, 2026 **Status:** CLOSED — 0.036% residual near-EXACT closure

Observation

PAPER_1209DD_S619 (Quantum/Atomic Unified Proof Set) gives the lead-digit form:

$$\text{Rydberg } R_\infty \approx R_\infty = F \cdot SO_5 + F \cdot SSQ + F^2 \cdot D_{phys} = 1.0974$$

Residual to CODATA: **0.036%**.

UQFF Closed Identity

$$R_\infty = F \cdot SO_5 + F \cdot SSQ + F^2 \cdot D_{phys} \approx 1.0974 \quad \text{residual } 0.036\%$$

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**: - **EM-coupling constants** (ϵ_0, μ_B, k_e) cluster around **$K_{MEX} \cdot D_{phys} = 8.33$** or **$N_{CH} = 9$** - **Planck/quantum constants** ($\hbar$) cluster around **$D_{BSFG} = 6$** - **Relativistic constants** (c) cluster around **$SO_5/D_{phys} = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Rydberg R_∞ as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S619
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "rydberg_r_inf_1_0974"})`

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